Final Project Review - TheFitWise

Manan Saini

IIIT Delhi, New Delhi, India

1 INTRODUCTION

In the contemporary health and wellness landscape, users encounter significant challenges in tracking their physical fitness, nutritional intake, and daily habits seamlessly across various isolated applications. The absence of a unified health assistant contributes to a disjointed user experience, adversely affecting consistency and goal attainment. In response to this issue, our all-encompassing web application, **TheFitWise**, serves as an indispensable daily health companion. It seamlessly incorporates features such as an AI-powered meal analyzer, comprehensive workout logging, gamified habit tracking, and integrated skincare routines, all consolidated into a single responsive dashboard.

2 MOTIVATION

Addressing the fragmentation in health tracking is not just a convenience but a critical enhancement to the overall user experience in personal wellness. Due to the lack of a cohesive tool, the issues faced are:

- **Fragmented User Experience:** The lack of a unified application leads to disjointed interactions. Users are forced to switch between separate apps for diet, workouts, and habit tracking, which hampers consistency.
- **Complex Meal Logging:** Current dietary applications often rely on tedious manual entry and barcode scanning, contributing to user frustration when logging complex or home-cooked meals.
- **Need for Personalized Assistance:** Users desire tailored support during their fitness journeys. The lack of an intelligent, context-aware assistant hinders their ability to calculate accurate calorie goals and receive on-the-fly wellness guidance.

3 TECHNOLOGIES & ALGORITHMS USED

- **Frontend Interface:** ReactJS 18 and Vite 5 are used to create an interactive, highly responsive single-page application (SPA). Tailwind CSS and Framer Motion provide a seamless and visually engaging UI.
- **Backend & Database:** Supabase is utilized for PostgreSQL database management, secure user authentication, and hosting serverless Edge Functions.
- **AI Integration:** The OpenAI API (gpt-4o-mini) is employed for natural language processing within the meal analysis module and the FitWise Chat widget.
- **Algorithms:** The Mifflin-St Jeor equation is implemented dynamically to auto-calculate daily caloric goals based on user profile metrics (age, weight, height, gender, and activity level).

4 NOVELTY

In comparison to existing systems, TheFitWise offers a significantly enhanced user experience through AI-driven natural language processing. A key feature is the **AI Meal Analysis**, which allows users to simply describe their meal in plain English (e.g., "2 boiled eggs and toast"), bypassing manual database searches.

Moreover, our approach stands out by unifying traditionally separate wellness domains. The inclusion of a dedicated Face & Skincare module—complete with an accurate built-in timer for facial exercises—alongside traditional gym tracking (50+ exercises) offers a more nuanced understanding of holistic health. Finally, the implementation of a gamified progression system fosters a more interactive and user-centric tracking experience.

5 RELATED WORK

Current products in the market, such as MyFitnessPal and Strong, serve as generalized tools for diet and workout tracking, respectively. However, they operate in silos. Previous works in diet tracking heavily rely on exact-match database queries, which struggle with unstructured input.

By leveraging LLMs for entity extraction and nutritional estimation, TheFitWise reduces the cognitive load required to maintain a food diary. Similarly, while traditional habit trackers introduce gamification, they lack deep integration with actual physiological data (weight trends, water intake, macro-nutrients). Our application bridges this gap by tying the XP and streak system directly to concrete health inputs.

6 PIPELINE & DATABASE ARCHITECTURE

The core backend architecture of TheFitWise is designed to process natural language into structured nutritional data while maintaining a robust relational database.

6.1 Database Schema The system relies on a PostgreSQL database hosted on Supabase, structured to handle complex user associations efficiently:

- profiles: Stores user settings, demographic data, and calculated calorie goals.
- meals & meal_items: Manages logged meals with granular, per-item macro breakdowns.
- workouts: Records session data, pre-built template usage, and quick check-ins.
- user_stats: The central hub for the gamification engine, tracking total XP, current level, active streaks, and unlocked badges.
- weight_logs, water_logs, caffeine_logs: Dedicated tables for daily quantitative health metrics.

6.2 AI Meal Analysis Pipeline

1. **User Input:** The user submits a plain text description of their meal via the frontend.
2. **Edge Function Invocation:** The request hits a secure Supabase Edge Function, protecting API keys and handling business logic.
3. **LLM Processing:** The Edge Function calls the OpenAI API (gpt-4o-mini) using a system prompt that instructs the model to act as a clinical nutritionist.
4. **Data Extraction:** The LLM identifies individual food items and estimates precise macronutrients returning structured JSON.

7 GAMIFICATION EVALUATION

7.1 Gamification Algorithms To drive user retention, a gamified habit system was engineered. The system awards Experience Points (XP) based on distinct user actions, such as completing a daily challenge, logging a meal, or hitting a water intake goal. Levels are calculated dynamically based on cumulative XP thresholds. Additionally, a streak tracking algorithm evaluates continuous daily logins, automatically issuing achievement badges (e.g., *Consistency King*) when specific milestones are reached.

7.2 Evaluation & Results The system was evaluated on frontend responsiveness and AI extraction accuracy.

- **Data Visualization:** The weight tracking module successfully computes and renders a 7-day rolling average using Recharts, effectively smoothing out daily fluctuations to reveal accurate long-term trends.
- **Performance:** The integration of TanStack React Query resulted in highly efficient client-side caching, minimizing redundant API calls to Supabase. The AI meal analyzer processed complex queries with an average response time of under 2 seconds, proving the viability of edge-function-based LLM integration for real-time user applications.

8 POTENTIAL CONTRIBUTIONS

Our contribution in this work is the development of a unified, open-source pipeline capable of transforming unstructured user health data into actionable, gamified insights. This project serves as a robust template for integrating modern React frontend ecosystems with serverless AI backends, making holistic health tracking more accessible and engaging for the end user.